PSE-TVD: Parameter Space Exploration for Time-Varying Ensemble Data Visualization and Analysis

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ABSTRACT

With the advance of supercomputers, scientists from multiple disciplines attempt to study more complex real-world phenomena with sophisticated computer simulation models. As the growing size of high-fidelity simulation data causes difficulties of storing and transferring, in situ visualization stands out as a scalable way for scientists to observe the data generated by their simulations. Although multiple image-based in situ visualization approaches have been proposed, those methods do not support parameter space exploration on time-varying simulation results.

We propose PSE-TVD, a deep learning based surrogate model to assist parameter space exploration (PSE) of time-varying ensemble data (TVD) visualization. PSE-TVD is designed to learn the mapping from the simulation parameters, time step, and viewpoints to the visualization results. Therefore, our approach allows parameter space exploration at different time steps of time-varying ensemble data without running the original time-consuming simulation or storing large size of simulation results for prediction. We evaluate our approach by qualitatively and quantitatively comparing to related approaches using multiple scientific datasets.

KEYWORDS

In situ visualization, time-varying data, parameter space exploration, generative adversarial network

1 INTRODUCTION

Scientific modelling has been playing an essential role in various fields. From the perspective of mathematics, a scientific model is the construction of mathematical relationships between observed natural phenomena. Many models simulate known phenomena using computers to make predictions or verify hypotheses, such as the process of nuclear explosions, weather predictions, and cosmology simulations.

In the real world, there are lots of uncertain factors that affect the environment. A single simulation result with a particular configuration is usually insufficient to verify the phenomena because of the inherent uncertainty. Therefore, scientists often try different simulation model configurations to demonstrate and compare the simulation results of an identical experiment. The output, which is a collection of spatio-temporal sequences, is referred to as an *ensemble* [39]. The simulation result for each configuration is called the *ensemble member*.

However, in the actual implementation, it faces the following two dilemmas:

- The data volumes of simulation outputs are enormous and not conducive to data transmission and post-hoc analysis.
- Ensemble simulation is time-consuming and cannot explore the parameter space effectively.

In situ visualization [1] generates visualization images on the device where simulation data is produced. We only need to save the visualization results rather than raw simulation data which is much more massive. Several works using in situ visualization were proposed to facilitate post-hoc analysis and exploration of parameter space. For example, the Cinema framework [30] enables visualization of the simulation data from different viewpoints in situ and supports post-hoc analysis. NPCP [40] visualizes the parameter of multi-resolution climate ensemble datasets and shows the correlation between the different convective parameters. However, these approaches have little consideration of the simulation parameters, which are important in studying ensemble simulations.

Surrogate models are designed to emulate the behavior of simulation models while alleviating the burden of simulation models' enormous time costs. Recently, a deep learning based surrogate models have been applied to parameter space exploration tasks. InsituNet [16] takes the simulation, visual mapping, and view parameters as input and shows the visualization results of ensemble simulations at short notice. DLA-VPS [35] takes the simulation parameters as input and generate the volumes of ensemble simulations. Nevertheless, the above surrogate models for parameter space exploration do not consider temporal information, which can only generate simulation results at the particular time step. Although the most straightforward method to derive time-series results using prior works is to train multiple models for multiple time steps, there are several drawbacks of such method: the training process of all models is time-consuming and the trained models take much storage space.

In this work, we propose PSE-TVD, a deep learning-based surrogate model to support the exploration of simulation parameters for time-varying ensemble simulations visualized in situ. That is, PSE-TVD enables the adjustment of time steps to derive the time-varying simulation results. Our work utilizes various image synthesis techniques, such as super resolution [34] and perceptual loss [4], to improve image prediction accuracy and fidelity. We train PSE-TVD to learn the end-to-end mapping from the simulation parameters, view parameters, and time step to visualization images. The trained model enables scientists to explore the time-series

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results and adjust the viewpoint for volume rendering without executing time-consuming simulations.

We demonstrate the effectiveness of PSE-TVD by comparing to the other related approaches using multiple scientific datasets. In summary, the contributions of this paper are as follows:

- A surrogate model (i.e., PSE-TVD) that can generate visualization images of time-varying ensemble simulation data.
- PSE-TVD supports adjustment of different viewpoints for visualizing the volume.

2 RELATED WORK

2.1 Image-based In Situ Visualization

In situ in computer science refers to bringing the computation to where data is located, substituting costly data transmission to facilitate post-hoc analysis. Our method belongs to image-based in situ visualization, which visualizes raw simulation data in situ and output images.

Ahrens et al. proposed Cinema Framework [30] that visualized the simulation data from different viewpoints in situ and collected rendered images for post-hoc analysis. Tikhonova et al. [36, 37] generated the images of multiple layers in situ to enable the adjustment of transfer functions for volume rendering. He et al. proposed InsituNet [16], a convolutional regression model to learn the mapping from the simulation and visualization parameters to the visualization results.

Our work can adjust simulation parameters, viewpoints, and time steps to generate the corresponding volume rendering image.

2.2 Deep Learning-based Surrogate Model

Surrogate model emulates the behavior of a simulation, generating approximate results of the original simulation for aiming to reduce the actual execution time and computing power of simulation.

There are some popular approaches to build a surrogate model: gradient-enhanced kriging [11, 12], radial basis function [31, 32, 41], support vector machines [5, 6], space mapping [22, 23], and artificial neural networks [3, 16].

More and more deep learning based surrogate models have been proposed in recent years. Hazarika et al. proposed NNVA [13], a fully connected neural network, which performs interactive parameter sensitivity analysis and estimates the uncertainty by dropout layers. Shi et al. proposed GNN-Surrogate [33], a graph neural network-based surrogate model to explore the parameter space of ocean climate simulations. He et al. proposed InsituNet [16], which is based on conditional GAN, enables adjustment of the simulation, visual mapping, and view parameters, and shows the approximate simulation result.

2.3 Deep Learning for Scientific Visualization

Recently, many researchers applied deep learning to visualization works. Hong et al.[18] used long short-term memory to estimate access pattern for parallel particle tracing. Han et al. used autoencoders to cluster streamlines and streamsurfaces [9].

Several works focused on facilitating ensemble datasets visualization and exploration of parameter space. Han et al. [10] used a recurrent generative network to perform the temporal super resolution for time-varying scientific datasets. Guo et al. [8] proposed a deep learning framework that produces the spatial super-resolution of vector field dataset. He et al. [15] used a convolution-based discriminative network to produce the collective comparison between two ensemble members.

We combine deep learning techniques and models on image synthesis (e.g., super resolution [34], self-modulation [4]) to enhance the quality and resolution of our volume rendering image results.

3 OVERVIEW

In this section, we describe the workflow of our approach. First, we collect training data by conducting ensemble simulations and perform in situ visualization. For each execution of simulation model, the data collection pipeline is shown in Figure 1. The simulation model takes a sampled set of simulation parameters as input and generate an ensemble member with T time steps. We sample N volumes from the member, visualizing them with the predefined transfer function and V sets of sampled view parameters. Three groups of parameters (i.e., simulation parameters, time step, and view parameters) and corresponding visualization image x constitute data pairs, which will be stored and used in training PSE-TVD. Furthermore, with the collected data, we train a deep learning model, PSE-TVD, to learn end-to-end mapping from parameters input to visualization output. Finally, the trained generator in PSE-TVD can be kept for predicting visualization images for arbitrary simulation, time step, and view parameters within the parameter space.

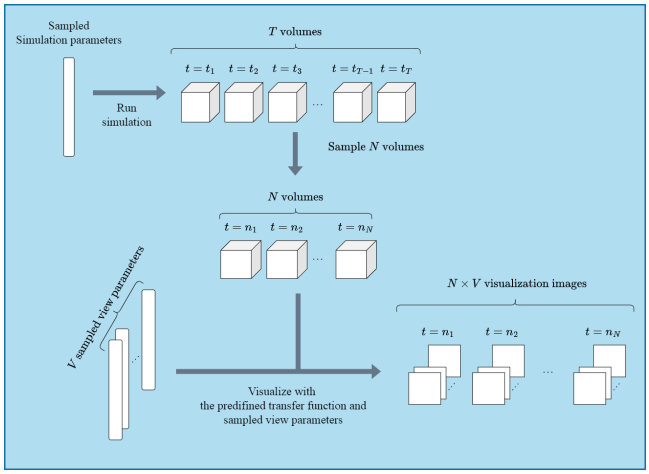


Figure 1: Visualization images collection pipeline for single execution of the simulation model.

In the following, we discuss three groups of parameters in detail.

Simulation Parameters P_{sim} are presented as a vector with one or more dimensions. The value range of each dimension is defined by domain scientists. By sampling simulation parameters within the defined ranges, the ensemble simulations are conducted to generate the ensemble data.

Time Step t is an index to represent the number of steps elapsed. To train a deep learning model that can predict visualization images

for arbitrary time steps, we sample volumes for each ensemble member and collect corresponding time steps for training.

View Parameters P_{view} are used in control the viewpoints for volume rendering. In this work, we define the viewpoints as angles of the camera rotating around the volume, which is controlled by azimuth $\theta \in [0, 360]$ and elevation $\phi \in [-90, 90]$. To train a deep learning model that can predict visualization images for arbitrary viewpoints, we sample sets of θ and ϕ, and generate images from the sampled viewpoints.

4 ARCHITECTURE AND TRAINING

PSE-TVD involves three subnetworks in the training stage, a generator G, a discriminator D, and a feature comparator F. The Generator G is a deep neural network learning the function that maps input parameters to visualization images. The discriminator D is a deep neural network learning to estimate the divergence between the distribution of predicted and ground truth images. The feature comparator F is a pretrained neural network that extracts image features (e.g., edges, shapes) for computing losses between predicted and ground truth images. After training, we only need the trained G to predict visualization images for arbitrary parameters within parameter space.

4.1 Generator

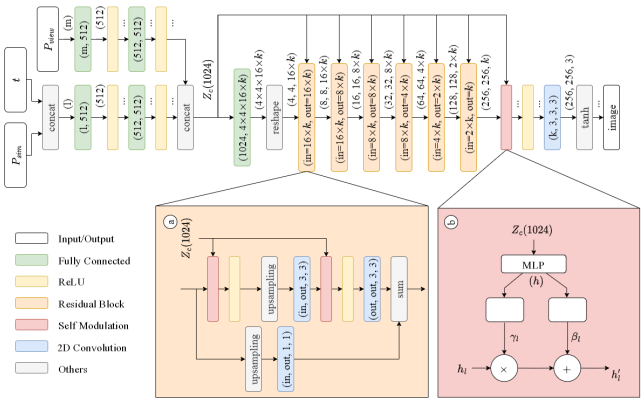


Figure 2: Generator

The architecture of generator G is shown in Figure 2. G takes P_{sim}, t, and P_{view} as inputs, produce the predicted image $\hat{x}$. First, P_{sim} and t, which are related to the volume, are concatenated and fed into fully connected layers. While P_{view}, which is related to visualization, is fed into another group of fully connected layers. The outputs are concatenated as latent vector Z_c, which is fed into all self-modulation layers. In addition, Z_c is encoded by another fully connected layer. The output is reshaped to a low-resolution image, which is enhanced to high-resolution images through residual blocks performing 2D convolution and upsampling. Note that we introduce constant k and h in the network architecture. k is the number of convolution kernels in immediate layers, and h is the hidden nodes in self-modulation layers. Both are used to balance the expressive power, architecture size, and training time of PSE-TVD to cope with datasets of different complexity.

Residual Block. The architecture of the residual block is shown in Figure 2a . For generating high-resolution images, the neural network layers could be very deep, which could cause gradient vanishing problems. We adopt residual blocks to improve the performance. For each residual block, the input image is upsampled by nearest neighbor interpolation, and the upsampled image is fed into two convolution layers with kernel size 3×3. In the end, the output is added to the original image. Note that to balance the dimension, we perform the upsampling and convolution with kernel size 1×1 to replace identity mapping.

Self-modulation. Batch normalization [20] is a widely used technique that improves the training of deep neural networks. The transformation is defined as:

$$h'_l = \gamma \odot \frac{h_l - \mu}{\sigma} + \beta \tag{1}$$

For layer l, h_l and h'_l are the input and output of l, respectively. μ and σ^2 are mean and variance computed over the dataset. γ and β are learned scale and shift parameters. However, for simulation data with different simulation parameters and time steps, the distribution of the volumes in the member is not the same. Using non-adaptive γ and β in l cannot map data properly. Self-modulation [4] , which replaces γ and β with input-dependent $\gamma_l(z)$ and $\beta_l(z)$, is shown in Figure 2b and defined as:

$$h'_l = \gamma_l(z) \odot \frac{h_l - \mu}{\sigma} + \beta_l(z) \tag{2}$$

$\gamma_l(z)$ and $\beta_l(z)$ are two modulated variables transformed from input z using feed forward network (MLP) with ReLU activation, separately.

4.2 Discriminator

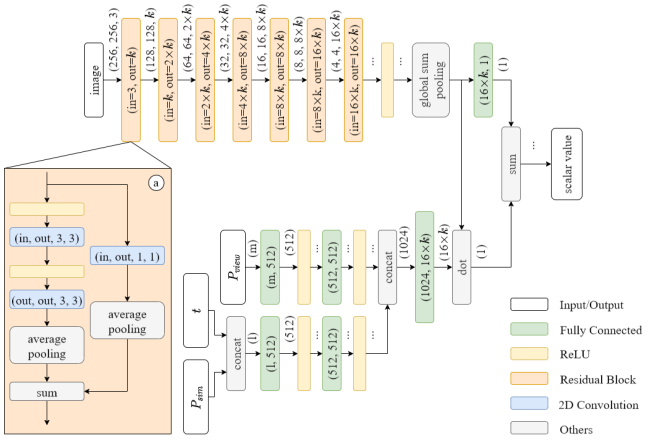


Figure 3: Discriminator

The architecture of D is shown in Figure 3, which takes a predicted/ground truth image and the corresponding parameters as inputs and produces a scalar value indicating how similar the input image is to the ground truth image conditioning in corresponding parameters. Similar to G, three groups of parameters are fed into fully connected layers, and the outputs are concatenated to

Z_c. Meanwhile, the input images are fed through residual blocks to extract the feature vector Z_f. Finally, the projection-based method [29] is used to integrate Z_c and Z_f into a scalar value. No batch normalization is performed because it often hurts the performance of D [24, 27].

Residual Block. The architecture of the residual block in D is similar to that in G except downsampling (average pooling in this work) is performed instead of upsampling to extract representations of features from images.

Spectral Normalization. Spectral normalization [28] is a weight normalization technique to enforce discriminator be Lipschitz continuous, stabilizing the adversarial training. We apply spectral normalization to all of the weights of the discriminator.

4.3 Feature Comparator

To synthesize high-quality images, we also adopt the pretrained VGG-19 model, a commonly used feature comparator, to extract certain types of image features. The extracted features from predicted and corresponding ground truth images are compared to minimize the difference between those features to improve the quality. According to [16], we use the layer $relu1_2$ for following two considerations:

- $relu1_2$ extracts low-level features such as edges and shapes, which commonly exist in scientific visualization images.
- The artifacts are introduced into generated image by pooling layers, $relu1_2$ is before the first pooling layer.

4.4 Loss Function

The training goal of G is minimizing the difference between the predicted image $\hat{x}$ using G and the corresponding ground truth image x. Inspired by previous works [16, 19, 21, 25], we define the loss function L_G by combining feature reconstruction loss [21] L_{F^l} and hinge loss [26, 38] L_{adv_G}:

$$L_G = L_{F^l} + \lambda L_{adv_G} \tag{3}$$

where λ is the coefficient to control the degree of L_{adv_G} involved in L_G. As suggested in [16], we set $\lambda = 0.01$ to balance the performance.

Feature Reconstruction Loss. The feature reconstruction loss between image $\hat{x}$ and x is defined by measuring the difference between their extracted features. Specifically, for a given image x, our feature comparator F (the pretrained VGG-19) is applied on x and extracts a set of feature maps, denoted as $F^l(x)$. Here l indicates which layer the feature maps are from ($relu1_2$ in this work). The extracted feature maps have the shape of $h \times w \times c$, where h, w, and c are the height, width, and number of channels, respectively. The feature reconstruction loss L_{F^l} between $\hat{x}$ and x can be defined as the pixel-wise mean squared error between $F^l(\hat{x})$ and $F^l(x)$:

$$L_{F^l} = \frac{1}{hwc}\mathbb{E}_{q(x),p(\hat{x})}[\parallel F^l(x) - F^l(\hat{x}) \parallel_2^2] \tag{4}$$

where q is the true distribution and p is the generator distribution.

Adversarial Loss. We also use adversarial loss of generator L_{adv_G} in L_G for measuring not only the difference between each pair of images with L_{F^l} but also the divergence between images distribution with L_{advG}.

In this work, we use hinge loss [26, 38], which is shown good performance when applying spectral normalization [28]. Let us denote the input image by x and the condition information (i.e., three groups of parameters) by y. Using q to designate the true distributions, the adversarial loss of generator L_{adv_G} and adversarial loss of discriminator L_{advD} are defined as:

$$L_{adv_G} = -\mathbb{E}_{q(y)}[D(G(y), y)] \tag{5}$$

$$\begin{aligned} L_{adv_D} =& \mathbb{E}_{q(y)}[\mathbb{E}_{q(x|y)}[\max(0, 1 - D(x, y))]] \\ &+ \mathbb{E}_{q(y)}[\max(0, 1 + D(G(y), y))] \end{aligned} \tag{6}$$

5 EXPERIMENT

In this section, we qualitatively and quantitatively evaluate the proposed PSE-TVD. We compare our approach with alternatives on different datasets, transfer functions, time steps, and viewpoints.

Dataset. We selected two datasets: Nyx and Deep Water Impact to carry out the evaluation.

Nyx is a cosmological simulation that solves equations of compressible hydrodynamics on an adaptive grid hierarchy coupled with an N-body treatment of the dark matter. Based on scientists interests, we study three parameters: total matter density ($OmM \in [0.17, 0.5]$), total density of baryons ($OmB \in [0.03, 0.08]$), and hubble constant ($h \in [0.55, 0.85]$). The simulation was conducted with 10 parameter settings sampled from the parameter space and generated the corresponding dark matter density field dataset with $64 \times 64 \times 64$ spatial resolution and 200 time steps.

Deep Water Impact is a dataset used to study asteroid impacts in deep ocean water from 2018 IEEE SciVis Contest. There are three paramter settings of the dataset: Airburst, Asteroid Diameter, and Entry Angle. Airburst represents four different airburst scenarios: None, 5km, 10km, and 15km elevation airburst, respectively. Asteroid Diameter represents three different basalt asteroid diameters: 100, 250 and 500 meter, respectively. Entry Angle represents three angles of entry: 27.4, 45 and 60 degrees off of the horizontal. The spatial resolution is $300 \times 300 \times 300$ and the temporal resolution is about 180 to 480 time steps. There are 11 scalar variables in the dataset, we use temperature variable in the experiment.

Tranfer function. In the context of the volume rendering, users can observe different features in the visualization if transfer functions change. This helps scientists view specific structures they are interested in. For each dataset, we perform volume rendering technique with two different transfer functions to generate the training images for our PSE-TVD model. We set all color maps in different transfer functions as "Blue-Green-Orange" defined in Paraview.

First transfer function (TF1): TF1 is aimed at showing overall structures of data. The scalar opacity function maps low scalar value to low opacity, and maps high scalar value to high opacity. The gradient opacity function maps low gradient to low opacity, and high gradient to high opacity.

Second transfer function (TF2): TF2 is aimed at showing the surfaces of data. The scalar opacity function is the same scalar opacity transfer function as TF1. The gradient opacity function maps low and high gradient to high opacity.

Sampling. The simulation model, denoted as S, takes a parameter setting x_i and generates a temporal sequence $S(x_i)$ (i.e., an

ensemble member m_i), consisting of T time steps, is defined as:

$$S(x_i) = m_i = \{s_1, s_2, ..., s_T\} \tag{7}$$

where s_i is the volume for the corresponding time step.

For each ensemble member m_i, we first sample N volumes $\hat{m}_i = \{\hat{s}_1, \hat{s}_2, ..., \hat{s}_N\}, \hat{s}_t \in m_i$. Then, we sample viewpoints of $\hat{s}_t$ with mapping Fibonacci lattice onto the surface of a surrounding sphere, and generate the image set in situ.

The resolution of visualized images are all 256×256 and the total number of visualized images are about 30000. After all training images are generated, we use 80% of them for training and 20% of them for testing.

Network architecture. For PSE-TVD, we set constants $k = 32$ and $h = 128$, which controls the number of convolutional kernels in the intermediate layers and the number of hidden nodes in self-modulation layers, separately. For InsituNet, We removed the visual mapping related parameters because we used a fixed transfer function without changing predefined operations to visualize the generated simulation data. Also, time step of corresponding visualization result is regarded as a parameter contained in simulation parameters. We set the constant $k = 35$ in generator and $k = 32$ in discriminator, which k controls the number of convolutional kernels in the intermediate layers. The total size of network architecture is not less than PSE-TVD.

Hyper-parameters. In the experiment, we trained all models with batch size 128, and set the exit criterion as the number of iterations to 46800.

All models are trained and tested on Taiwan Computing Cloud, which contains 2 NVIDIA V100 GPUs with NVlink.

5.1 Quantitative Evaluation

For quantitative evaluations, we used four metrics that focus on different aspects to compare the predicted images with the ground truth images, including peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), earth mover's distance (EMD) between color histograms [2], and Frechet inception distance (FID) [17].

PSNR represents the ratio between the maximum possible power of a signal and the power of corrupting noise. It measures difference between two images using aggregated mean squared error. A higher PSNR indicates that the compared images are similar in pixel domain.

SSIM measures the similarity between two images based on degradation of structural information. A higher SSIM means the compared images are similar from the structural point of view.

EMD is used in [2] to quantify the distance between color histograms of two images. A lower EMD means the compared images have similar color distribution.

FID approximates the distance between two distributions of images. A lower FID means two images are similar statistically.

The quantitative results are shown in Table 1. PSE-TVD gives all best metrics on Nyx with all transfer functions and on Deep Water Impact with TF2. On Deep Water Impact with TF1, PSE-TVD gives best metrics EMD and FID while PSNR and SSIM are slightly lower than those of InsituNet.

The training time of PSE-TVD is all shorter than InsituNet , however, the forward propagation time of PSE-TVD is slower than InsituNet on all datasets with all transfer functions.

5.2 Qualitative Evaluation

For each transfer function using on each dataset, we selected two time steps to compare the ground truth images with the images generated from PSE-TVD, and InsituNet.

Figure 4 shows Deep Water Impact dataset visualized with different transfer functions. Compared to InsituNet, our approach offers a much more details when TF1 is used to visualize the data (Figure 4 (a)). While using TF2 (Figure 4 (b)), InsituNet generates apparent false contours. Figure 5 shows the Nyx dataset with different transfer functions. Compared to InsituNet, Our approach shows higher accuracy of features and is much similar to the ground truth image. As illustrated in Figure 5 (b), InsituNet generates a noticeable artifacts in the images.

Table 2 shows the storage sizes of visualization images with different transfer functions and trained generator in different models. With the smaller architecture size than InsituNet, we showed that PSE-TVD provides better mapping from input parameters to corresponding visualization results quantitatively and qualitatively.

6 CONCLUSION AND FUTURE WORK

In this work, we present PSE-TVD, a new deep learning based surrogate model for assisting parameter space exploration of time-varying ensemble data. Our model regards the time step as the parameter related to the corresponding volume, learning the mapping from simulation parameters, time step, and view parameters to visualization result. Additionally, to handle the simulation results with different distributions, we adopt self-modulation, which is an architectural modification providing better mapping results. In future work, we will extend our work to directly predict the corresponding volumes instead of visualization images because volumes are more feasible for scientists to analyze.

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Dataset	Transfer Function	Model	Metric				Timing	
			PSNR↑	SSIM↑	EMD↓	FID↓	t_{tr} (hr)	t_{fp} (s)
Nyx	TF1	InsituNet	43.688	0.980	0.0012	12.809	11.32	**0.012**
		PSE-TVD	**46.005**	**0.985**	**0.0010**	**3.472**	**10.51**	0.020
	TF2	InsituNet	32.593	0.928	0.0051	35.075	11.31	**0.009**
		PSE-TVD	**40.824**	**0.971**	**0.0015**	**4.612**	**10.45**	0.022
Deep Water Impact	TF1	InsituNet	**39.917**	**0.971**	**0.0012**	39.259	11.01	**0.013**
		PSE-TVD	38.162	0.958	**0.0012**	**10.835**	**10.08**	0.019
	TF2	InsituNet	34.876	**0.941**	0.0021	31.250	11.00	**0.010**
		PSE-TVD	**35.475**	**0.941**	**0.0014**	**14.242**	**10.07**	0.020

Table 1: Metrics and Timings: The quantitative results from datasets with different transfer functions. t_{tr} is timing for training the model and t_{fp} is timing for a forward propagation of the trained model.

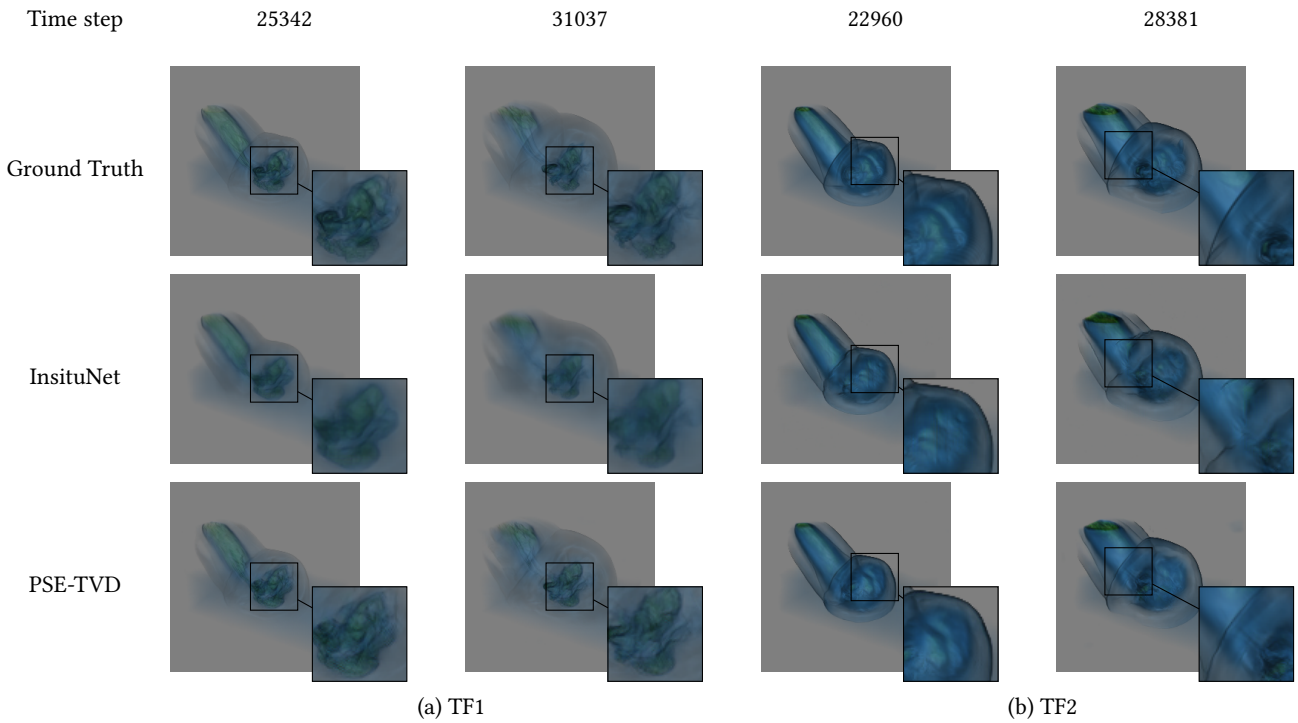


Figure 4: Comparison of the images generated using InsituNet and PSE-TVD with the ground truth images of Deep Water Impact dataset. Images in the first two columns are visualized results with TF1 at 25342 and 31037 time step. Images in the last two columns are visualized results with TF2 at 22960 and 28381 time step.

Dataset	Model	Size (MB)		
		TF1	TF2	Net size
Nyx	InsituNet	926	1126	77.03
	PSE-TVD			76.94
Deep Water Impact	InsituNet	698	766	77.03
	PSE-TVD			76.94

Table 2: The sizes of images generated with different transfer functions and trained models.

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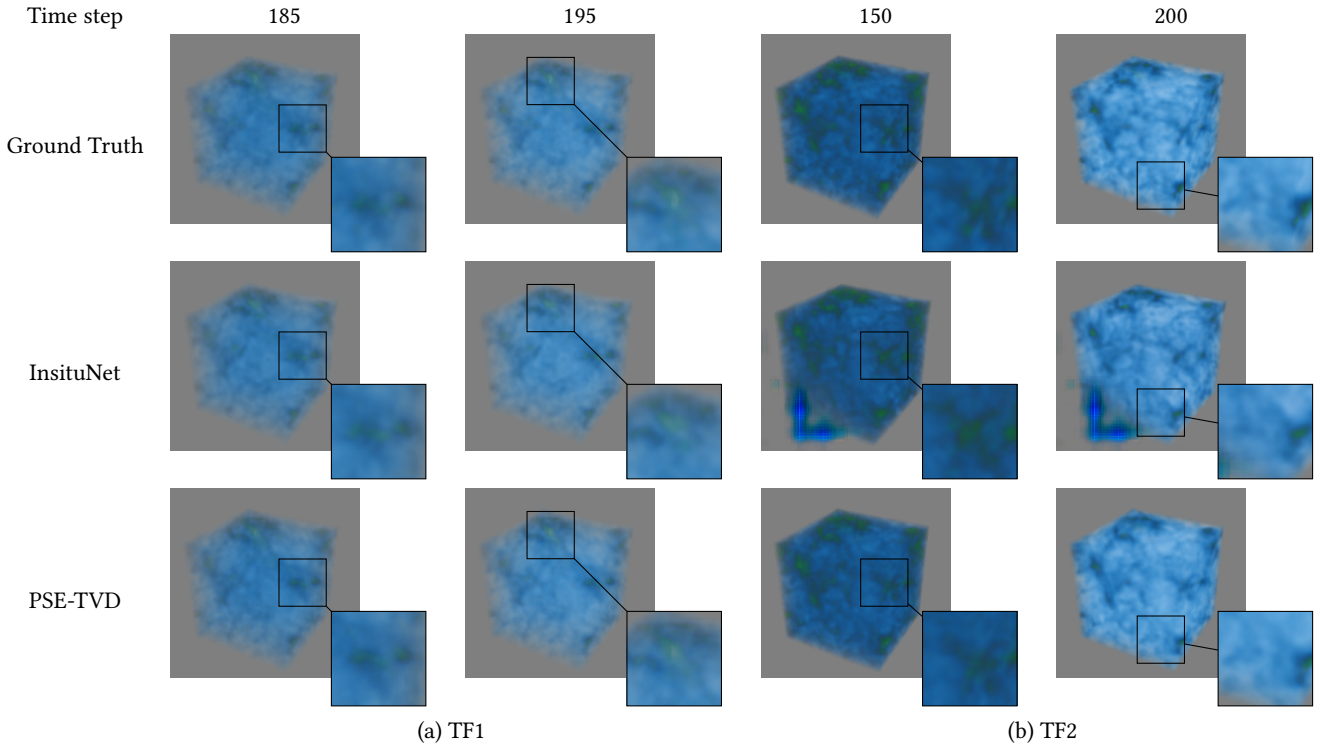


Figure 5: Comparison of the images generated using InsituNet and PSE-TVD with the ground truth images of Nyx dataset. Images in the first two columns are visualized results with TF1 at 185 and 195 time step. Images in the last two columns are visualized results with TF2 at 150 and 200 time step.

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